

MarketSense Project Report

Generated on 2026-01-04

1. Week One: Market Data Ingestion

Objective: Build a robust system to ingest real-time stock market data.

Implementation Details & Fetching Mechanism:

- Platform: Yahoo Finance API (accessed via `yfinance` wrapper).
- Method: The system executes a `download()` call requesting the last 730 days of 1-day interval data for the batch of tickers.
- Real-Time Logic: The script runs in a continuous `while True` loop. Every 15 minutes (configurable), it:
 1. Re-establishes a session with Yahoo Finance.
 2. Downloads the freshest OHLCV candle for the current timestamp.
 3. Updates the internal DataFrame and recalculates indicators instantly.

Why it is Best:

- Cost-Effective: Uses free, high-quality data from Yahoo Finance.
- Comprehensive: Doesn't just fetch price; it calculates deep technical signals instantly.
- Reliability: Handles API connection errors and duplicates automatically.

2. Week Two: News & Sentiment Analysis

Objective: Integrate qualitative data (News) to complement the quantitative data (Price).

Implementation Details & Fetching Mechanism:

- Platform: Google News RSS Feed.
- Method: We construct a dynamic URL query for each stock: `https://news.google.com/rss/search?q={{TICKER}} AND (site:reuters.com OR site:bloomberg.com...)`.
- Real-Time Logic: The script polling loop runs every 5 minutes:
 1. Iterates through the watchlist (`AAPL`, `NVDA`, etc.).
 2. Sends an HTTP GET request to the Google RSS endpoint for that specific ticker.
 3. Parses the XML response using `feedparser` to extract the latest <item> tags (headlines).
 4. Immediately passes text to VADER for sub-second sentiment scoring.

Why it is Best:

- Targeted: Filters noise by only allowing trusted financial publishers.
- Actionable: Converts complex text into a simple 'Traffic Light' signal (Green/Red/Grey).
- Speed: Lightweight VADER model runs locally with zero latency.

3. How it Works (Real-Time Architecture)

The system achieves 'Real-Time' results through a high-frequency polling architecture:

1. Shared Configuration: Both pipelines read from `watchlist.txt`, ensuring they always track the user's defined assets.
2. Independent Loops:
 - The Market Data Loop polls Yahoo Finance for price changes.
 - The News Loop polls Google RSS for information changes.

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3. Instant Processing: As soon as data arrives (Price or Text), it is processed in-memory (0.01s latency) to generate indicators/verdicts, providing the user with an up-to-the-second view of the market state.

4. Input & Output Samples

Input Command (Terminal):

```
python news_ingestion.py --interval 60
```

Output Sample (Terminal):

```
INFO:sentiment_analysis: [VADER Sentiment Analyzer initialized]
[Loaded Watchlist]: AAPL, NVDA, TSLA, MSFT, GOOGL

=====
MARKET SENSE UPDATE | 23:28:56
=====
INFO:__main__: [Fetching news for $AAPL...]

AAPL Analysis
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Sentiment: +0.19 (Positive)
Verdict:   [BUY]
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1. [Financial Times] EU readies tougher tech enforcement in 2026 as Trump warns of retaliation
2. [Financial Times] US to extend productivity lead on back of AI boom, say economists
3. [CNBC] These are BTIG Research's top stock picks for 2026
4. [Bloomberg] How to Save the US from Authoritarianism
5. [Financial Times] Twitter and Pinterest founders launch app as antidote to social media

INFO:__main__: [Fetching news for $TSLA...]

TSLA Analysis
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Sentiment: -0.17 (Negative)
Verdict:   [SELL]
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1. [The Wall Street Journal] Stocks to Watch Friday: Tesla, Baidu, Micron, Wayfair
2. [Reuters] Berlin power grid attack caused by 'extreme leftists', officials say
3. [Reuters] Venezuela: Maduro in NY custody, Caracas defiance and oil
4. [Bloomberg] Activist Group Claims Responsibility for Berlin Power Outage
5. [Bloomberg] Oil Market May Absorb Maduro Shock Amid Abundant Global Supplies
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